

Supplementary material

Long-chain α -olefins production over Co-MnO_x catalyst with optimized interface

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1. Experimental methods

1.1 Experimental details of the FTS tests and product analyses

The schematic diagram of the reaction system for the FTS experiments along with the online analysis devices is given in Fig. SE1. The FTS reactions were performed in a high-pressure stainless-steel fixed-bed reactor with an inner diameter of 9 mm. Typically, 0.5 g catalyst (40-60 mesh) diluted with an equal volume of quartz sands (40-60 mesh) was loaded at the isothermal zone of the reactor. Before the FTS test, the catalyst was reduced at 400 °C and 0.1 MPa for 5 h in a pure H₂ flow of 50 mL·min⁻¹. After the reduction, the reactor was cooled to 120 °C in the same H₂ flow. Then, the syngas with the volumetric composition of 32% CO, 64% H₂, and 4% Ar as the internal standard was switched, and pressurized to 1.0 MPa. The catalytic test was conducted at the reactor temperature range from 210 to 230 °C and gas hourly space velocity (GHSV) from 2.4 to 9.6 L·g_{cat}⁻¹·h⁻¹. The tail gas of the reactor was depressurized before entering an online gas chromatograph (GC), and the entire pipeline was insulated at 300 °C to avoid the product condensation. The heated flow was firstly introduced to a HP-PONA capillary column via a six-way valve, and the hydrocarbons (C₁–C₁₈) were separated and analyzed using an FID. The representative chromatograms are given in Fig. SE2 and SE3. The pipelines, valves, and chromatographic injection port were insulated at 300 °C, and the temperature of the HP-PONA capillary column was increased from 25 to 280 °C at a ramp of 10 °C/min and held at 280 °C for 20 minutes. The flow out of the six-way valve was cooled in a cold trap (0 °C) to collect liquid products. The CO, CH₄, Ar, and CO₂ gases in the effluent from the cold trap were separated using the PROPAC Q and CST-TDX-01 packed columns at 50 °C and analyzed on a thermal conductivity detector (TCD), and the typical chromatograms are shown in Fig. SE4. All the catalytic results were obtained at a time on stream of no less than 24 h and each catalyst was tested at least for three times for calculating the experimental errors. After the reaction, the reactor was purged with N₂ (200 mL/min) for 15 min and cooled

naturally in the N₂ flow to the room temperature (20-25 °C). Then the catalyst was passivated for 1 h in a 1 vol % O₂/Ar flow of 20 mL/min before unloading from the reactor. The unloaded spent catalyst was sealed in the polyester sample tube after eliminating the air in a glove box under an Ar protection.

The catalytic activity is expressed as the CO conversion, cobalt-time yield (CTY) and turnover frequency (TOF), respectively. The CO conversion (X_{CO}) is defined according to Eq. S1, where $F_{\text{CO,in}}$ and $F_{\text{CO,out}}$ are the molar flow rate of CO at the inlet and outlet of the reactor, respectively.

CTY is defined as moles of CO converted to hydrocarbons over per gram of cobalt per second as expressed in Eq. S2, where m_{Co} is the mass of cobalt in the reactor.

TOF is defined according to Eq. S3, where the “moles of exposed Co⁰ atoms” is calculated based on the H₂-chemisorption.

The selectivity of hydrocarbon (S_{CxHy}) is defined according to Eq. S4, where F_{CxHy} represents the molar flow rate of the hydrocarbon at the outlet of the reactor.

To reveal the accuracy of the online GC analyses and determine the carbon balance of the FTS experiment, longer test (TOS>100 h) was performed over Co-MnO_x-(C)/SiO₂ catalyst under the conditions of 210 °C and 7.2 L·g_{cat}⁻¹·h⁻¹ with CO conversion of ~10%. The products were collected in the cold trap under the steady state (after a TOS of ~24 h). After statically leaving the sample tube for

about 1 h, the products were spontaneously separated into an aqueous phase and an oil phase. To determine the amount of oxygenate products, the aqueous phase sample was dissolved in acetone (HPLC grade, Sigma Aldrich) and analyzed on an offline GC (Shimadzu 2010) equipped with a capillary column (DB-5, 30 m × 0.32 mm) and an FID. From the representative chromatogram (Fig. SE5), C₁₋₄ alcohols were detected and their molar concentrations in the aqueous phase were determined by the external standard method. The selectivity of oxygenate (S_{oxy.}) was calculated according to Eq. S5, where $n_{oxy.}$ represents the moles of carbon in the oxygenates within the aqueous phase, and $n_{CO,consumed}$ represents the moles of consumed CO during the product collecting process. The selectivity of C₁₋₄ alcohols was determined to be less than 2%. This is consistent with the common fact that oxygenates are much less produced for the Co-based catalysts under FTS conditions with a low pressure.

$$S_{oxy.} = \frac{n_{oxy.}}{n_{CO,consumed}} \times 100\% (Eq. S5)$$

After a strong blending, a small amount of the oil sample was dissolved in 15.0 mL CS₂ (A.R. grade, Aladdin). The hydrocarbons dissolved in CS₂ were subjected to a full analysis on an offline GC (Agilent 7890B) equipped with a capillary column (RIPP GSD-1, 10 m × 530 μm × 0.15 μm), FID, and an analysis software of the simulated distillation method (Provided by the Sinopec Research Institute of Petroleum Processing Co., Ltd.). The temperature of the column furnace was programmed to rise from 50 to 430 °C at a temperature ramp of 10 °C/min and hold at 430 °C for 20 minutes. The representative FID signals of the oil phase are displayed in Fig. SE6. Since that the selectivity of C₇ hydrocarbons calculated based on the online GC analyses is reasonably the same to that based on the offline GC analyses of the oil products, the selectivity of hydrocarbons in the collected oil is determined. Results indicate that the discrepancy for the selectivity of C₅-C₁₈ hydrocarbons between the online and offline GC analyses is below ± 3%. To confirm the accuracy, the FTS reaction was also operated for a higher CO conversion of ~30% under the conditions of 220

121 °C and $4.8 \text{ L} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$. The representative FID signals of the oil phase are displayed in Fig. SE7. The
 122 selectivity of C_{1-4} alcohols is still below 2% and the difference in the selectivity of $\text{C}_5\text{-C}_{18}$
 123 hydrocarbons determined based on the online and offline GC analyses is still within $\pm 3\%$. Thus, the
 124 online GC analyses of the hydrocarbons with a carbon number up to 18 are practically accurate
 125 within the experimental errors.

126 Based on the FTS results with the full analyses of the condensed products, the carbon balances
 127 were also estimated according to Eq. S6. The carbon balance is defined as the total amount of carbon
 128 in all of the products divided by the amount of converted CO. Since that the selectivity of FTS
 129 products is calculated based on carbon number, the carbon balance can also be expressed as the sum
 130 of the selectivity of CO_2 , $\text{C}_1\text{-C}_5$ hydrocarbons based on the online GC analyses, and C_{6+} hydrocarbons
 131 and the oxygenates based on offline GC analyses. From the FTS results obtained under the two sets
 132 of conditions, i.e., 210°C and $7.2 \text{ L} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$ vs. 220°C and $4.8 \text{ L} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$, the carbon balances are
 133 reproducibly determined to be within $100 \pm 10\%$. Thus, the experimental errors of the FTS reaction
 134 used in our case are accurate within $100 \pm 10\%$ of the carbon balances.

$$135 \quad C_{\text{balance}} = \frac{n_{\text{CO}_2} + n_{\text{HC}} + n_{\text{oxy.}}}{n_{\text{CO, consumed}}} \times 100\%$$

$$136 \quad S_{\text{CO}_2, \text{online}} + S_{\text{C1-C5, online}} + S_{\text{C6+, offline}} + S_{\text{oxy., offline}} \quad (\text{Eq. S6})$$

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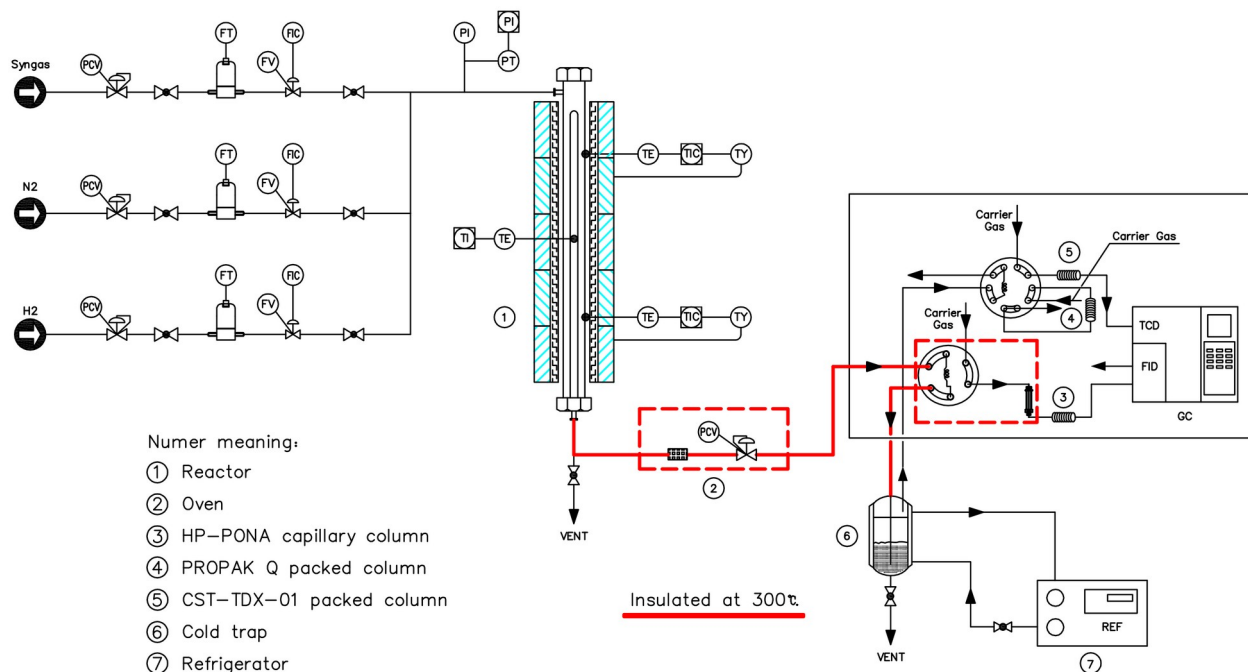
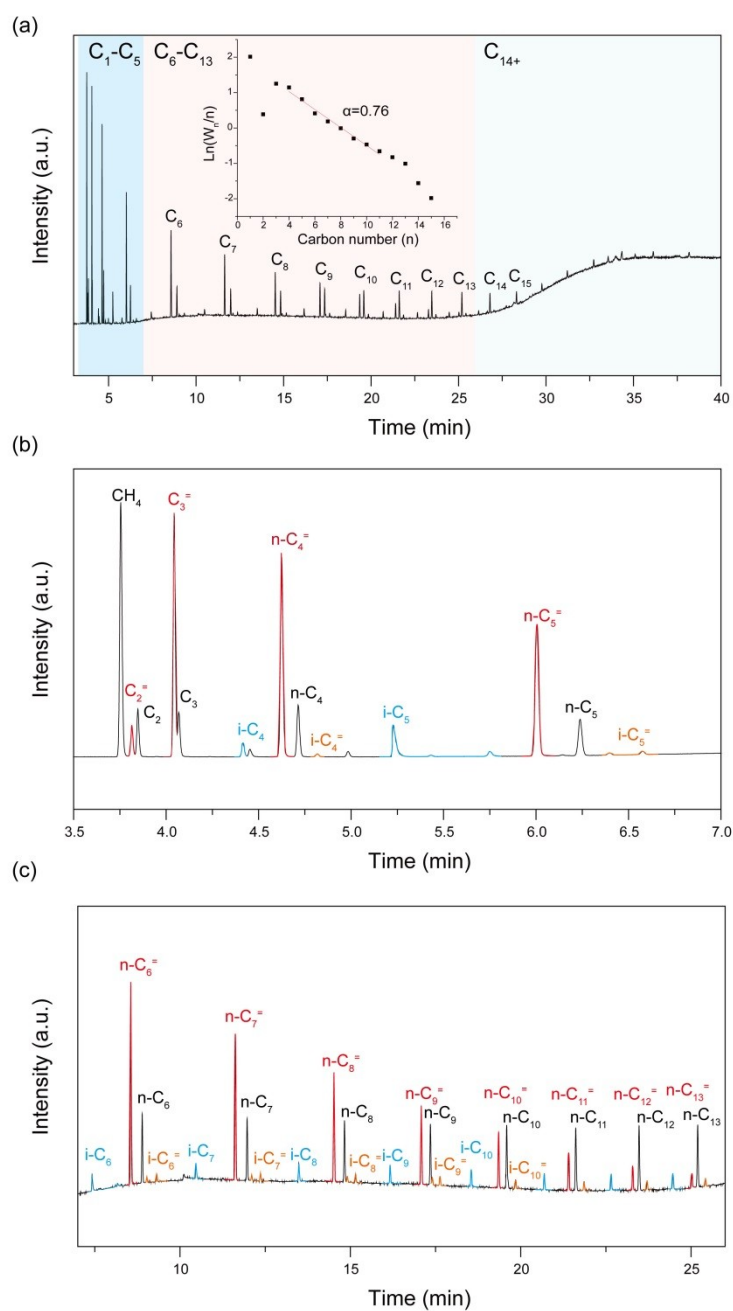
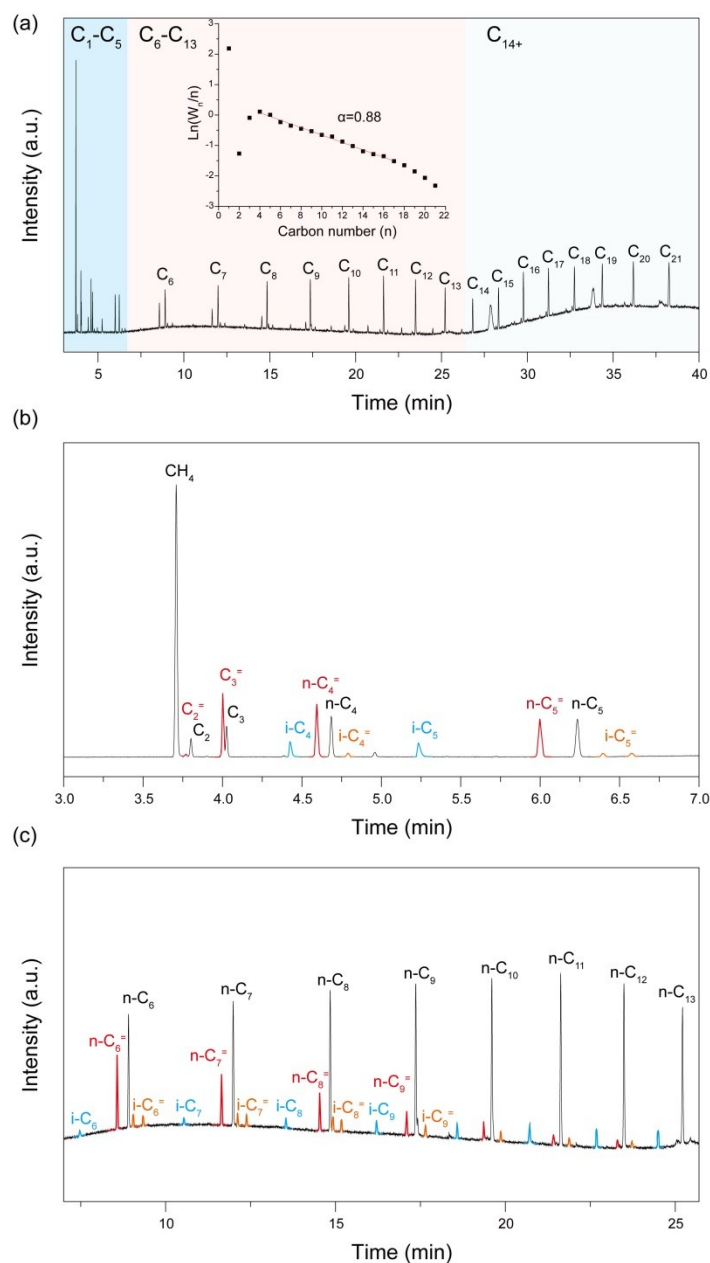


Fig. SE1. Schematic diagram of the FTS reaction system including the online GC analysis.



141
 142 **Fig. SE2.** Chromatogram of hydrocarbon products from a HP-PONA capillary column. (a) FID signals of a
 143 representative online hydrocarbon products from a fixed-bed FTS catalytic test of Co-MnO_x-(C)/SiO₂ at CO conversion
 144 of ~10%. Inset: ASF plot and the chain growth probability ($\alpha = 0.76$) obtained in the C₃-C₁₆ hydrocarbon range. Detail
 145 view of hydrocarbons: (b) C₁-C₅, and (c) C₆-C₁₃. Color code: black (*n*-paraffin products, *n*-C_{*n*}), red (*n*-olefin products, *n*-
 146 C_{*n*}⁻), blue (isomeric paraffin products, *i*-C_{*n*}), orange (isomeric olefin products, *i*-C_{*n*}⁻). Reaction conditions: 210 °C, 1.0
 147 MPa, GHSV=7.2 L·g_{cat}⁻¹·h⁻¹, H₂/CO = 2 and the data is collected after more than 24 h on stream.
 148



149

150 **Fig. SE3.** Chromatogram of hydrocarbon products from a HP-PONA capillary column. **(a)** FID signals of a
 151 representative online hydrocarbon products from a fixed-bed FTS catalytic test of Co-(C)/SiO₂ at CO conversion of
 152 ~10%. Inset: ASF plot and the chain growth probability ($\alpha = 0.88$) obtained in the C₃-C₁₆ hydrocarbon range. Detail
 153 views of hydrocarbons: **(b)** C₁-C₅, and **(c)** C₆-C₁₃. Color code: black (*n*-paraffin products, *n*-C_n), red (*n*-olefin products, *n*-
 154 C_n⁼), blue (isomeric paraffin products, *i*-C_n), orange (isomeric olefin products, *i*-C_n⁼). Reaction conditions: 210 °C, 1.0
 155 MPa, GHSV=4.8 L·g_{cat}⁻¹·h⁻¹, H₂/CO = 2 and the data is collected after more than 24 h on stream.

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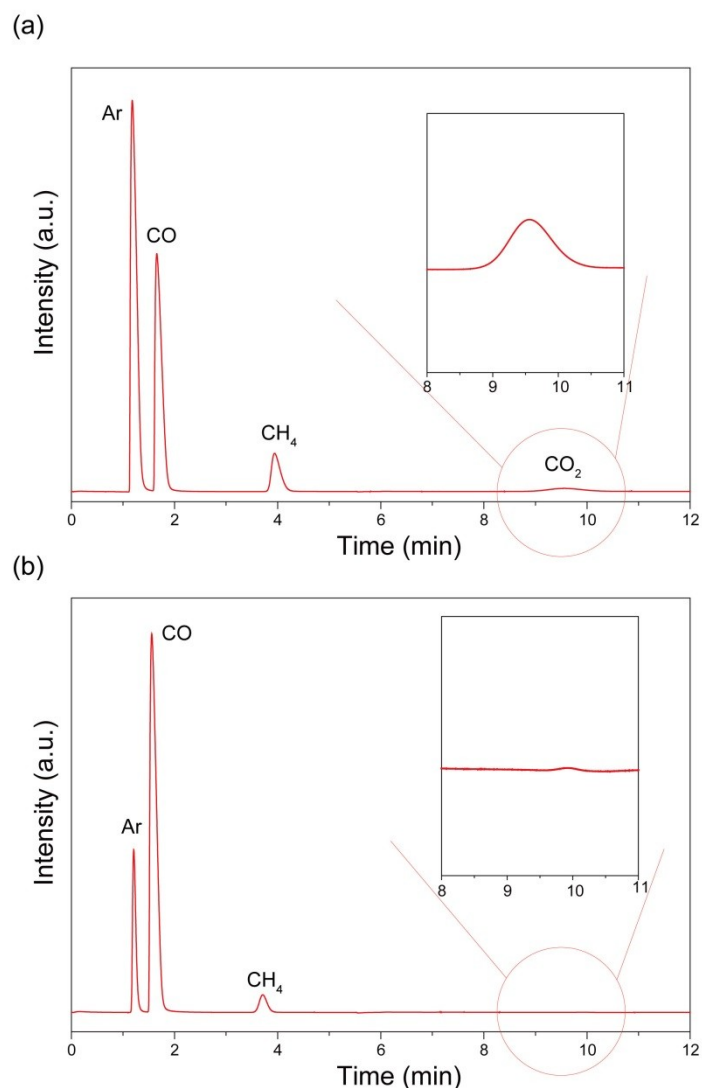
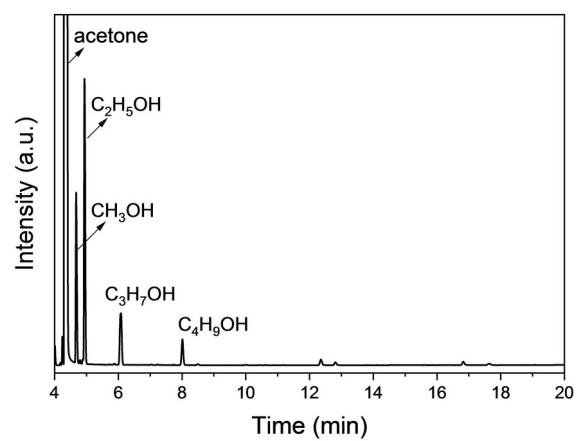


Fig. SE4. TCD signals of permanent gases eluting from the PROPAK Q and CST-TDX-01 packed columns. **(a)** TCD signals of the standard mixed gases of Ar (30%), CO (20%), CH₄ (5%), and CO₂ (2%). **(b)** TCD signals of a representative online test of Co-MnO_x-(C)/SiO₂ at CO conversion of ~40%. The invisible CO₂ signal suggests the negligible CO₂ selectivity (<1%). Reaction conditions: 220 °C, 1.0 MPa, GHSV = 4.8 L·g_{cat}⁻¹·h⁻¹, H₂/CO =2.



163

164 **Fig. SE5.** The representative chromatogram of the aqueous phase products obtained from the offline GC analysis.

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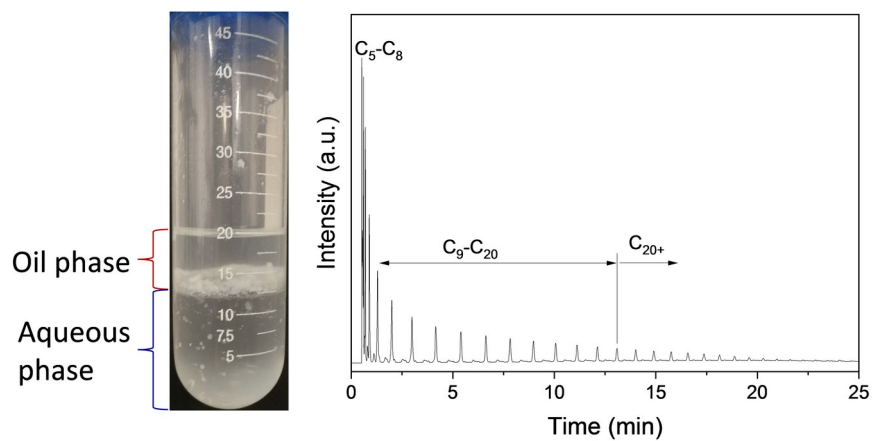
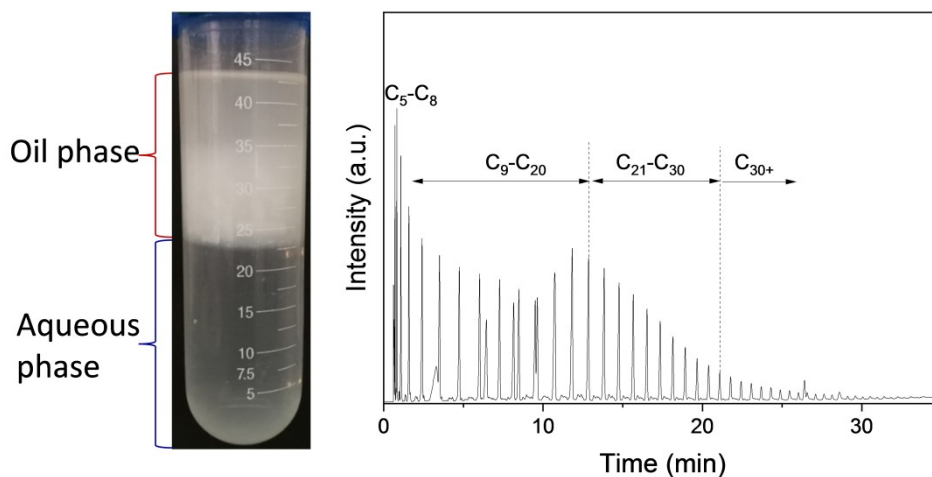


Fig. SE6. FTS products collected from a stability test over $\text{Co-MnO}_x\text{-(C)/SiO}_2$ at $210\text{ }^\circ\text{C}$, $7.2\text{ L}\cdot\text{g}_{\text{cat}}^{-1}\cdot\text{h}^{-1}$ at the quasi steady-state conditions (after 24 h on-stream) (left). The FID signals of the oil phase products obtained from the offline GC analysis (right).



171

172 **Fig. SE7.** FTS products collected from a stability test over Co-MnO_x(C)/SiO₂ at 220 °C, 4.8 L·g_{cat}⁻¹·h⁻¹ at the quasi
 173 steady-state conditions (after 24 h on-stream) (left). The FID signals of the oil phase products obtained from the offline
 174 GC analysis (right).

175

1.2 Mass transport limitation

The influence of the interphase diffusion limitation of reactants on the tested catalysts with particle diameter of 350 μm was estimated based on the reaction results in Table S2 (220 $^{\circ}\text{C}$) and Table S3 (210 $^{\circ}\text{C}$). The occurrence of internal diffusion limitation was determined according to the Weisz–Prater criterion (Eq. S7) [1, 2], where if C_{WP} (dimensionless parameter) is less than 1, internal diffusion limitation is absent.

r_{A} : reaction rate ($\text{mol}_{\text{CO}} \cdot \text{kg}_{\text{Co}}^{-1} \cdot \text{s}^{-1}$).

ρ_{Co} : Co density over the catalyst ($\text{kg}_{\text{Co}} \cdot \text{m}_{\text{particle}}^{-3}$).

R_{c} : average radius of the catalyst particle.

ε_{p} : porosity of the catalyst bed.

$D_{\text{CO, wax}}$: diffusion coefficient of CO in wax [3].

$H_{\text{CO, wax}}$: Henry coefficient, $2 \times 10^{-2} \text{ MPa} \cdot \text{m}^3 \cdot \text{mol}^{-1}$.

P_{CO} : CO partial pressure, 0.32 MPa.

Table SE1. The related parameters estimated by Weisz–Prater criterion

Catalyst	$-r_{\text{A}}$ ($\times 10^{-2} \text{ mol}_{\text{CO}} \cdot \text{kg}_{\text{Co}}^{-1} \cdot \text{s}^{-1}$)	ρ_{Co} ($\text{kg}_{\text{Co}} \cdot \text{m}_{\text{particle}}^{-3}$)	$R_{\text{c}}^{[\text{c}]}$ ($\times 10^{-4} \text{ m}$)	$D_{\text{CO, wax}}$ ($10^{-8} \text{ m}^2 \text{ s}^{-1}$)	C_{wp}
Co/SiO ₂ ^[a]	1.7	46.4	1.75	1.5	0.06
Co-MnO _x /SiO ₂ ^[a]	1.5	46.4	1.75	1.5	0.05
Co-(C)/SiO ₂ ^[a]	1.6	46.4	1.75	1.5	0.06
Co-MnO _x -(C)/SiO ₂ ^[a]	2.6	46.4	1.75	1.5	0.09
Co/SiO ₂ ^[b]	2.1	46.4	1.75	1.7	0.07
Co-MnO _x /SiO ₂ ^[b]	2.6	46.4	1.75	1.7	0.09
Co-(C)/SiO ₂ ^[b]	2.4	46.4	1.75	1.7	0.08
Co-MnO _x -(C)/SiO ₂ ^[b]	5.6	46.4	1.75	1.7	0.18

^[a] The reaction data is acquired from Table S2 under the FTS reaction condition of 210 $^{\circ}\text{C}$, 1.0 MPa.

^[b] The reaction data is acquired from Table S3 under the FTS reaction condition of 220 $^{\circ}\text{C}$, 1.0 MPa.

^[c] The granular-sizes of catalysts (40-60 mesh) are about 350 μm , thus average radius of 175 μm are used to calculate

194 C_{wp} .

195 According to the calculated effectiveness factors listed in Table SE1, C_{wp} value is substantially
196 less than 1 for the four tested catalysts under the reaction conditions of 210 °C with CO conversion
197 of $10 \pm 1.4\%$, implying no diffusion limitations of the reactants on the surface reactions. Therefore,
198 the measured intrinsic activity can be safely correlated with the chemical and structural properties of
199 catalysts. When increasing reaction temperature to 220 °C, the internal diffusion limitation still can
200 be ignored in the cases of Co/SiO₂, Co-MnO_x/SiO₂, and Co-(C)/SiO₂ with CO conversion less than
201 35%. But for the catalyst of Co-MnO_x-(C)/SiO₂ with extremely high CO conversion of ~80% at 220
202 °C, the relatively high C_{wp} value suggests some extent of the internal diffusion limitation on the rates
203 of surface reactions.

204 In order to estimate the influence of external diffusion on the catalyst performance of Co-MnO_x-
205 (C)/SiO₂ at 210 °C, the CO conversions were sequentially measured by increasing the reactant flow
206 rate (volume of reactant per unit time) and catalyst volume (V) while keeping their ratio (space
207 velocity, WHSV) constant [4]. The CO conversion and hydrocarbon product selectivity as function
208 of flow rate were plotted and shown in Fig. SE8. Note that the flow rate has no influence on the CO
209 conversion and product selectivity, as well as the chain growth probability, suggesting the external
210 mass diffusion limitation is negligible for Co-MnO_x-(C)/SiO₂ with CO conversion less than 15% at
211 210 °C, 1.0 MPa.

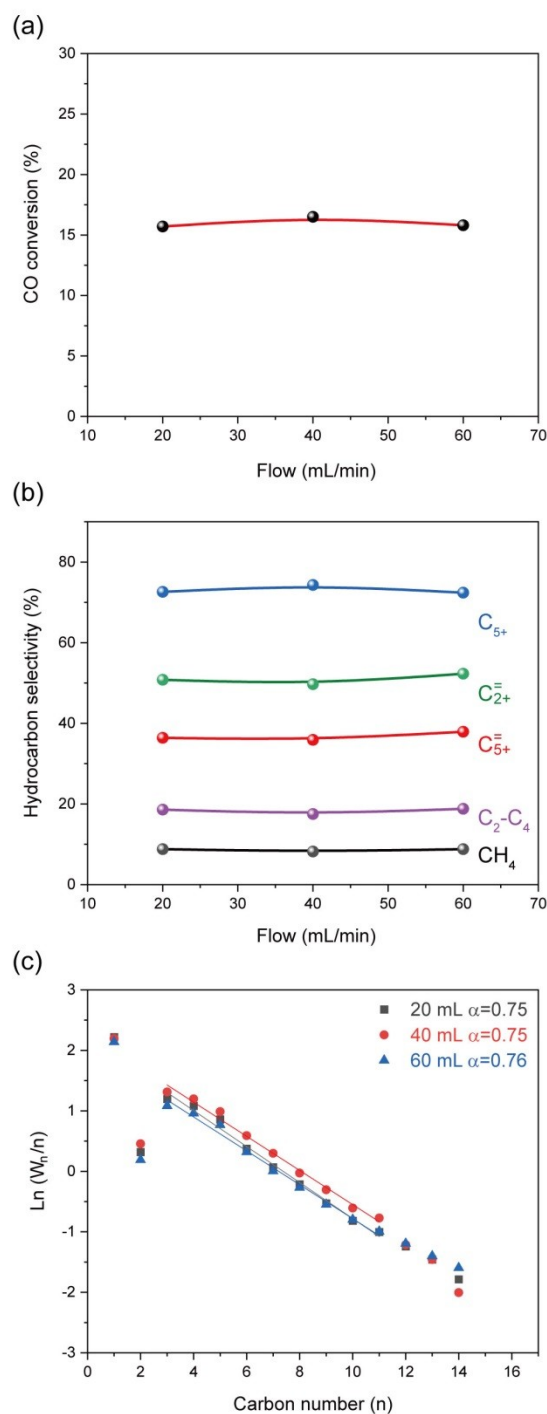


Fig. SE8. Flow rate test of Co-MnO_x-(C)/SiO₂ by keeping WHSV constant. The influences of flow rate on (a) CO conversion and (b) hydrocarbon product selectivity. (c) ASF plots with the corresponding chain growth probabilities in the C₃-C₁₁ hydrocarbon range derived from online product analysis. Reaction conditions: 210 °C, 1.0 MPa, H₂/CO = 2, and the data is collected after more than 24 h on stream. Cat. = 0.25 g for 20 mL/min, 0.50 g for 40 mL/min, and 0.75 g for 60 mL/min.

219 **2. Supplementary Figures**

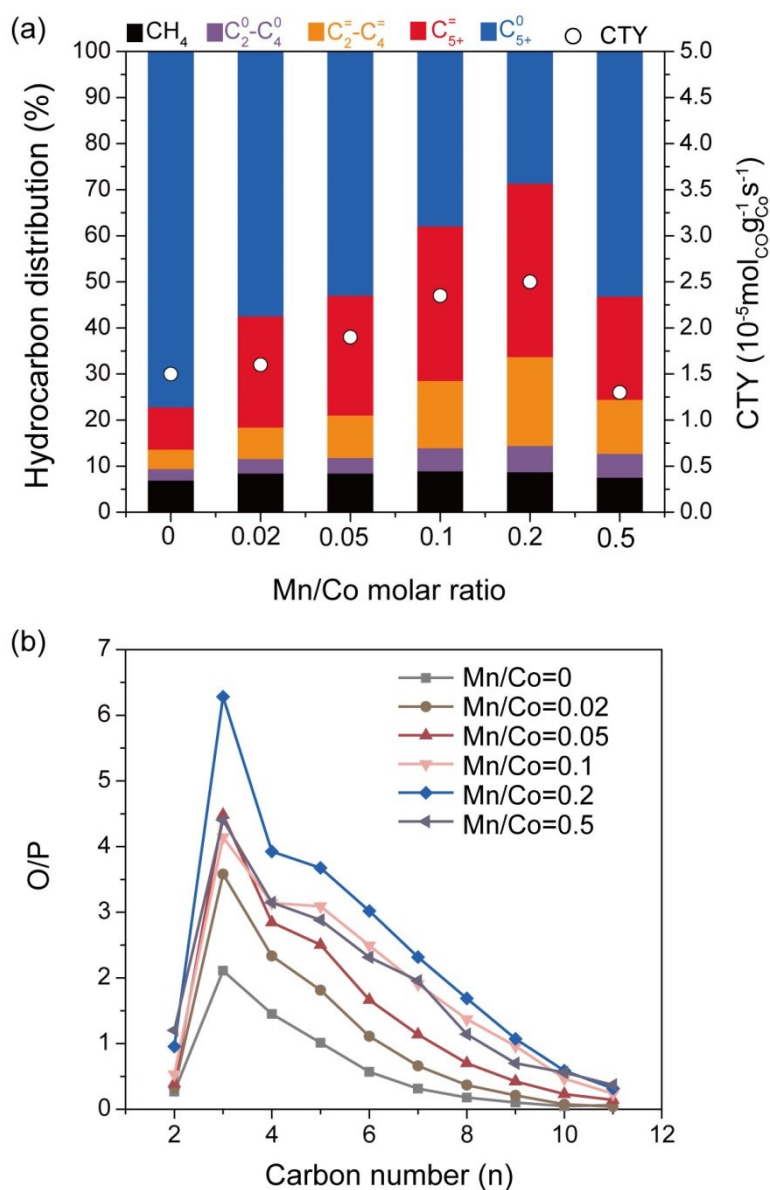


Fig. S1. FTS results over Co-MnO_x-(C)/SiO₂ with different Mn/Co molar ratios. (a) CTY and product distributions. (b) Olefin/paraffin (O/P) ratio of the products. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 210 °C, H₂/CO = 2.0, GHSV = 4.8 L·g_{cat}⁻¹·h⁻¹, and the data was collected after more than 24 h on stream.

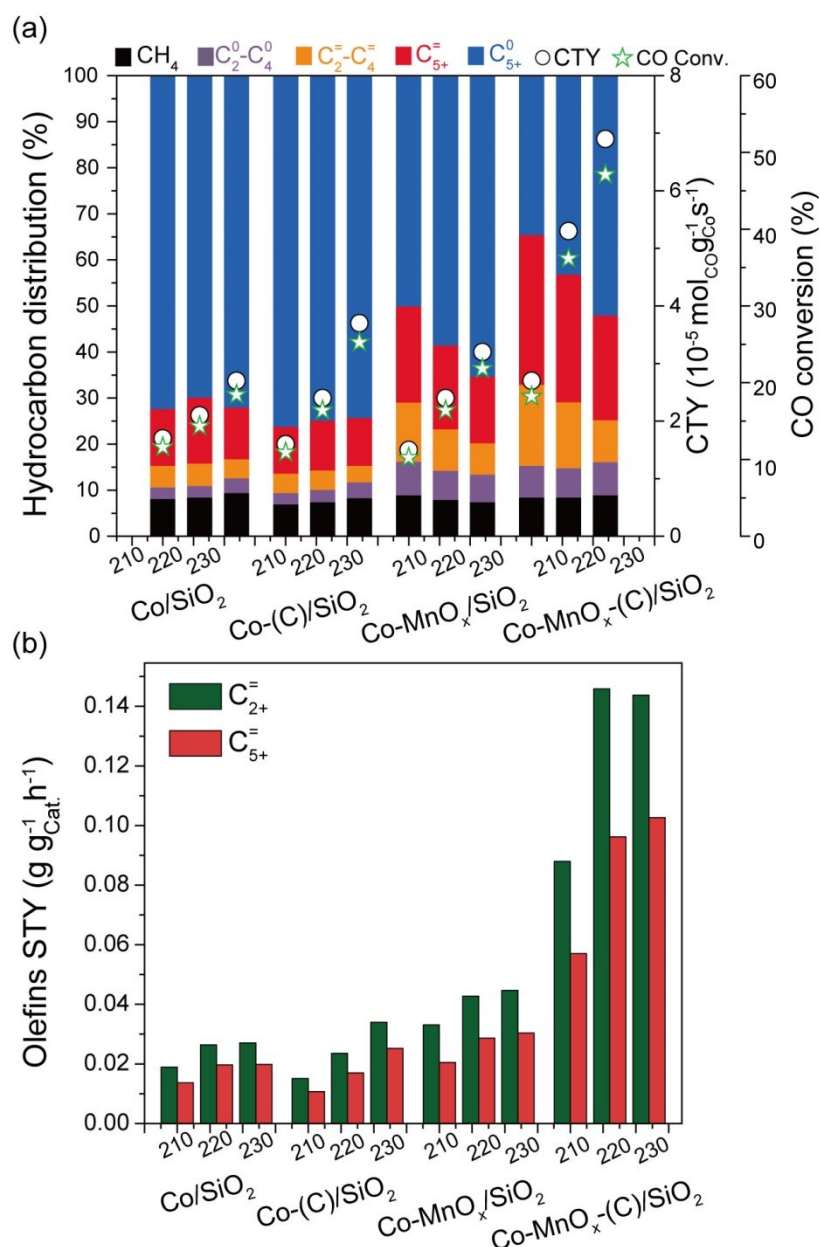


Fig. S2. FTS results of the four catalysts at different reaction temperatures. (a) CTY, CO conversion and product distributions. (b) Space-time yield (STY) of C₂+ and C₅+. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 4.8 L·g_{cat}⁻¹·h⁻¹, H₂/CO = 2.0, and the data was collected after more than 24 h on stream.

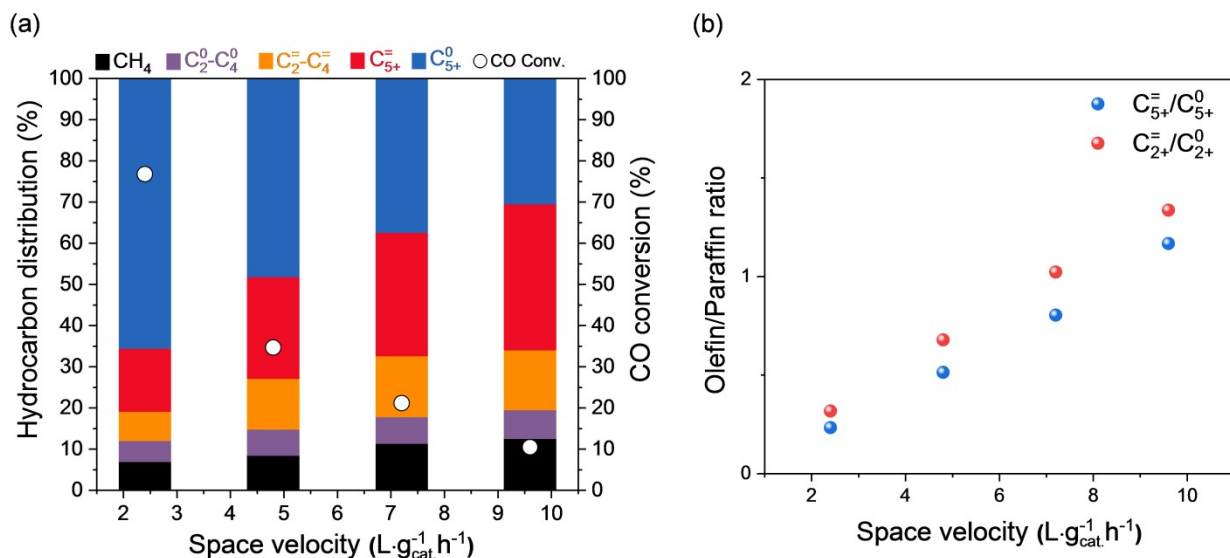


Fig. S3. (a) CO conversion and product distribution, and (b) O/P ratio of C₂₊ and C₅₊ over Co-MnO_x-(C)/SiO₂ at different GHSV. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 220 °C, GHSV = 2.4 ~ 9.6 L·g_{cat}⁻¹·h⁻¹, H₂/CO = 2.0, and the data was collected after 24 h on stream.

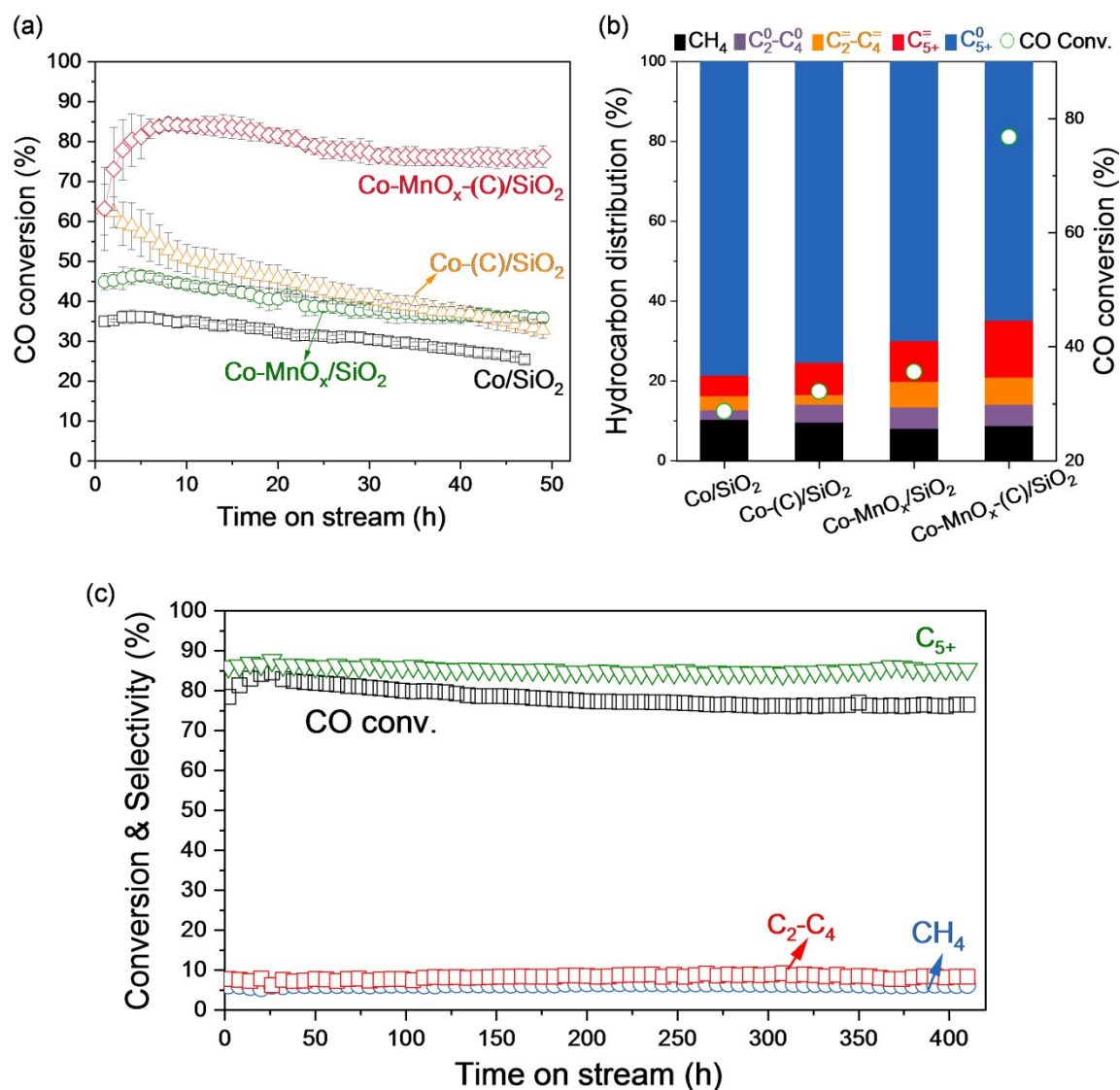


Fig. S4. Catalyst stability tests, (a) CO conversion as a function of the time-on-stream for the catalysts. (b) CO conversion and product distribution at the stable stage. (c) Long-term stability results over $\text{Co-MnO}_x\text{-(C)/SiO}_2$. Reaction conditions: 0.50 g catalyst, 1.0 MPa, 220 °C, $\text{H}_2/\text{CO} = 2.0$, and $\text{GHSV} = 2.4 \text{ L} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{h}^{-1}$.

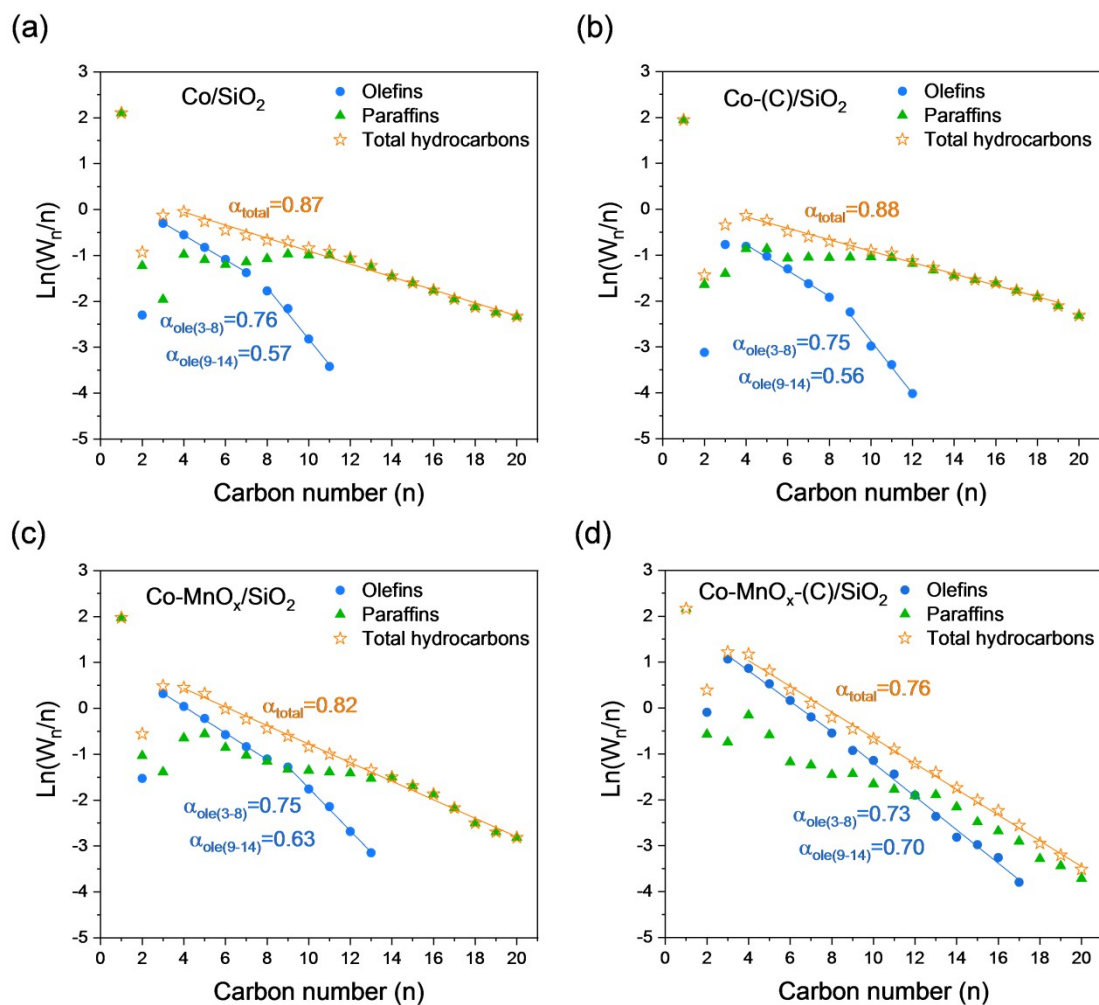
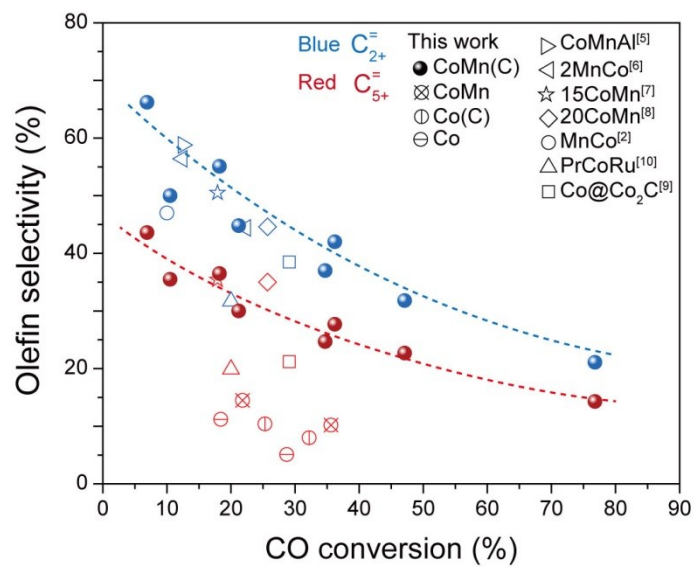


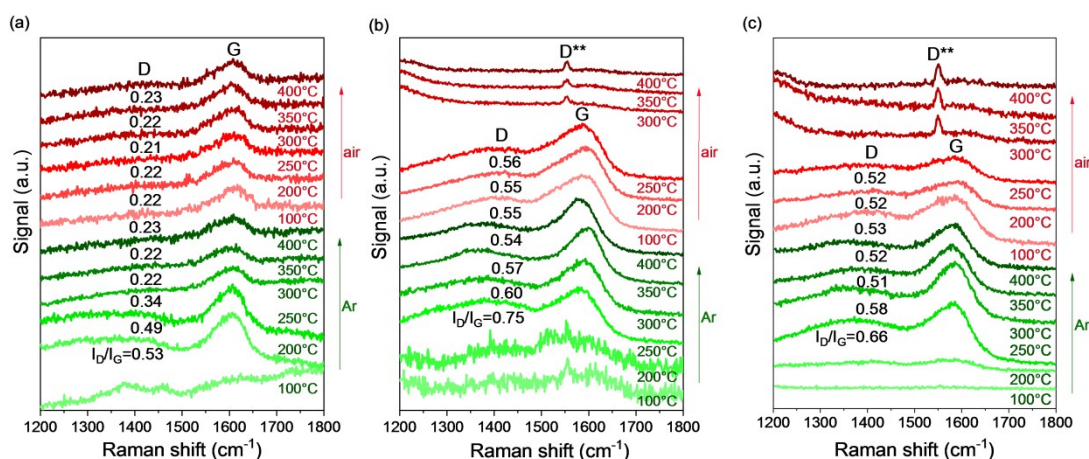
Fig. S5. ASF distribution of olefins and total hydrocarbons. **(a)** Co/SiO₂, **(b)** Co-(C)/SiO₂, **(c)** Co-MnO_x/SiO₂, and **(d)** Co-MnO_x-(C)/SiO₂. Reaction conditions: 0.50 g catalyst, 210 °C, 1.0 MPa, H₂/CO = 2.0, GHSV of 4.8 to 7.2 L·g_{cat}⁻¹·h⁻¹ for achieving the CO conversion at 10.0 ± 1.4%.



246

247 **Fig. S6.** The dependences of olefin selectivity on the CO conversion over Co-MnO_x-(C)/SiO₂ related catalysts and those
248 reported in the references.

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250

251 **Fig. S7.** *In situ* Raman spectra of the dried samples of (a) Glu/SiO₂, (b) Co-Glu/SiO₂, and (c) Co-Mn-Glu/SiO₂ during the
 252 consecutive treatments in the flow of Ar and air with increasing temperature.

253 In order to monitor the evolution of carbon species during heating process, the dried samples of SiO₂ impregnated with
 254 glucose (Glu/SiO₂), co-impregnation of Co(NO₃)₂ and glucose (Co-Glu/SiO₂), and co-impregnation of Co(NO₃)₂,
 255 Mn(CH₃COO)₂ and glucose (Co-Mn-Glu/SiO₂) were tested by *in situ* Raman technique during the consecutive treatments in
 256 a flow of Ar and air with increasing temperature. Specifically, 50 mg sample was loaded in the *in situ* cell (Harrick, HVC-
 257 DR2) and heated in a flow of Ar (20 mL/min) at atmospheric pressure with stepwise increasing temperature to 400 °C (5
 258 °C/min). The corresponding spectrum at the 1200-1800 cm⁻¹ region was recorded at 100, 200, 250, 300, 350 and 400 °C,
 259 respectively. After cooled down to room temperature, the sample was switched to a flow of air (20 mL/min) and heated
 260 again with stepwise increasing temperature to 400 °C (5 °C/min). The corresponding spectrum at the 1200-1800 cm⁻¹ region
 261 was recorded at 100, 200, 250, 300, 350 and 400 °C, respectively.

262

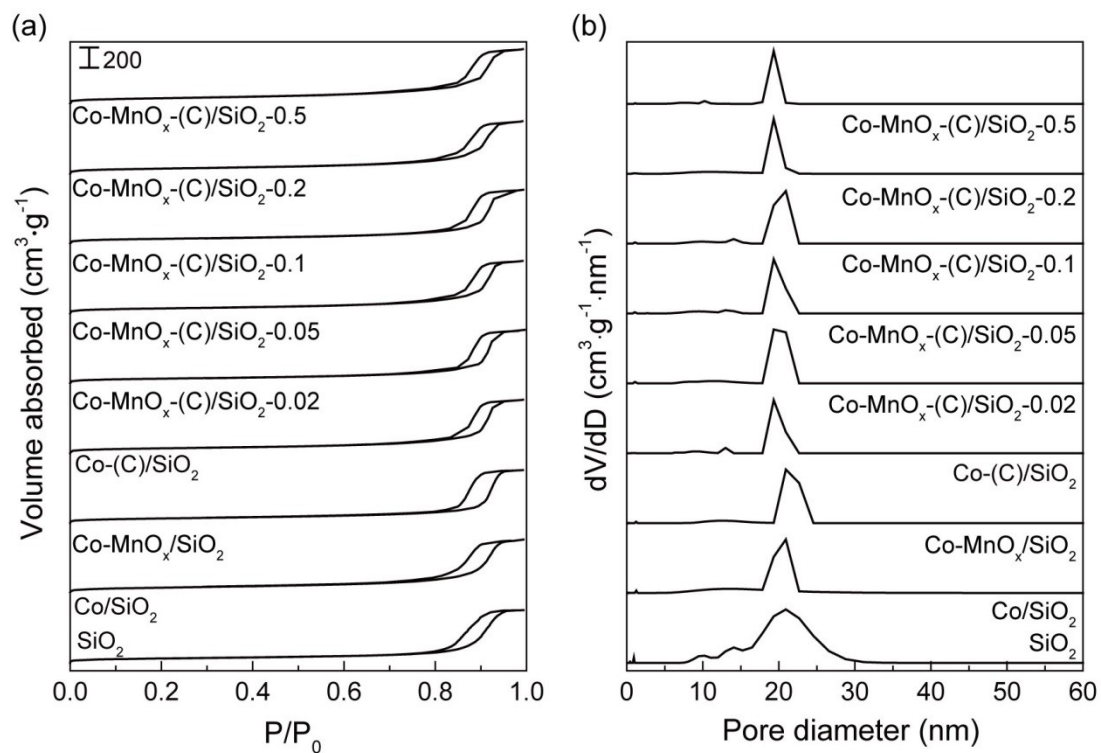


Fig. S8. Textural properties of the calcined catalysts. **(a)** N_2 adsorption-desorption isotherms. **(b)** Pore size distributions calculated by BJH mesopore model.

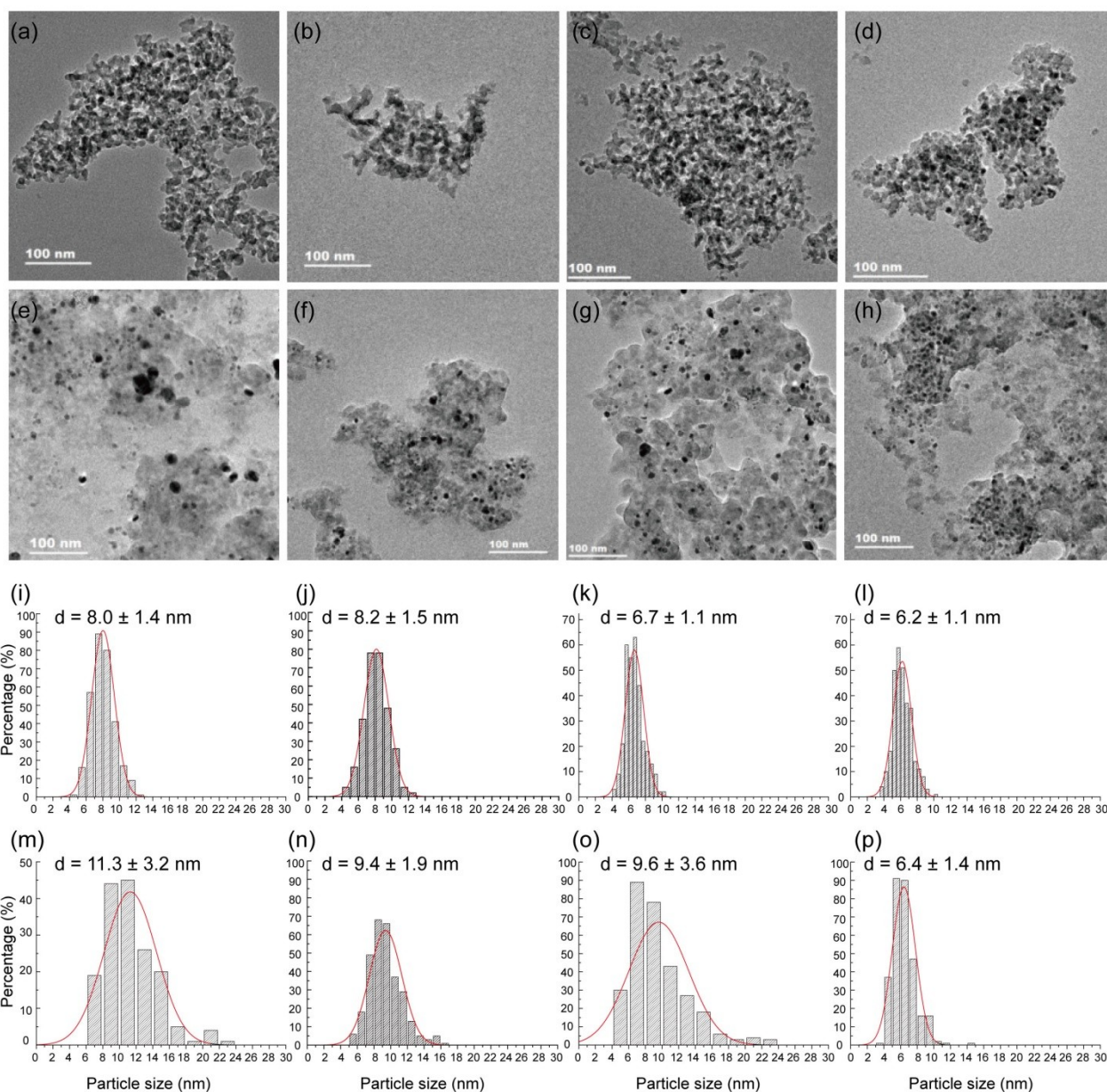


Fig. S9. TEM images, particle size distributions, and the average particle sizes of the calcined and used catalysts. Calcined catalysts: (a, i) Co/SiO₂, (b, j) Co-MnO_x/SiO₂, (c, k) Co-(C)/SiO₂, and (d, l) Co-MnO_x-(C)/SiO₂. Used catalysts: (e, m) Co/SiO₂, (f, n) Co-MnO_x/SiO₂, (g, o) Co-(C)/SiO₂, and (h, p) Co-MnO_x-(C)/SiO₂.

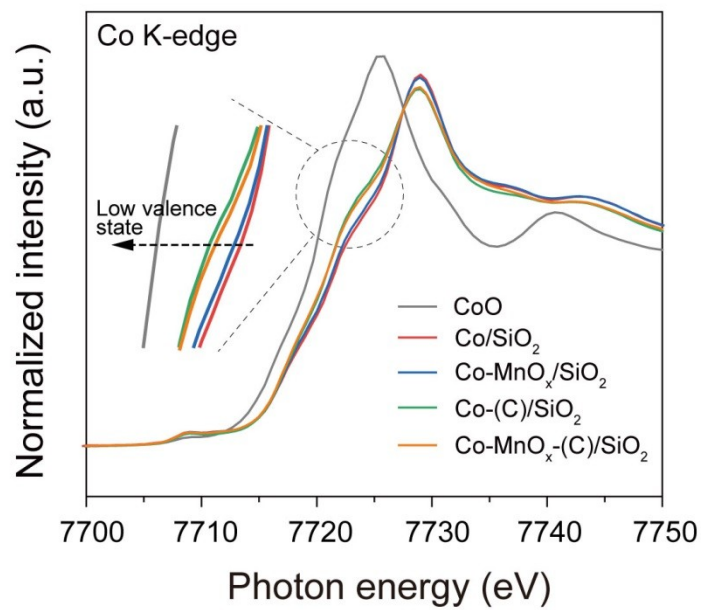


Fig. S10. Normalized XANES spectra of calcined catalysts at the Co K-edge with the standard CoO.

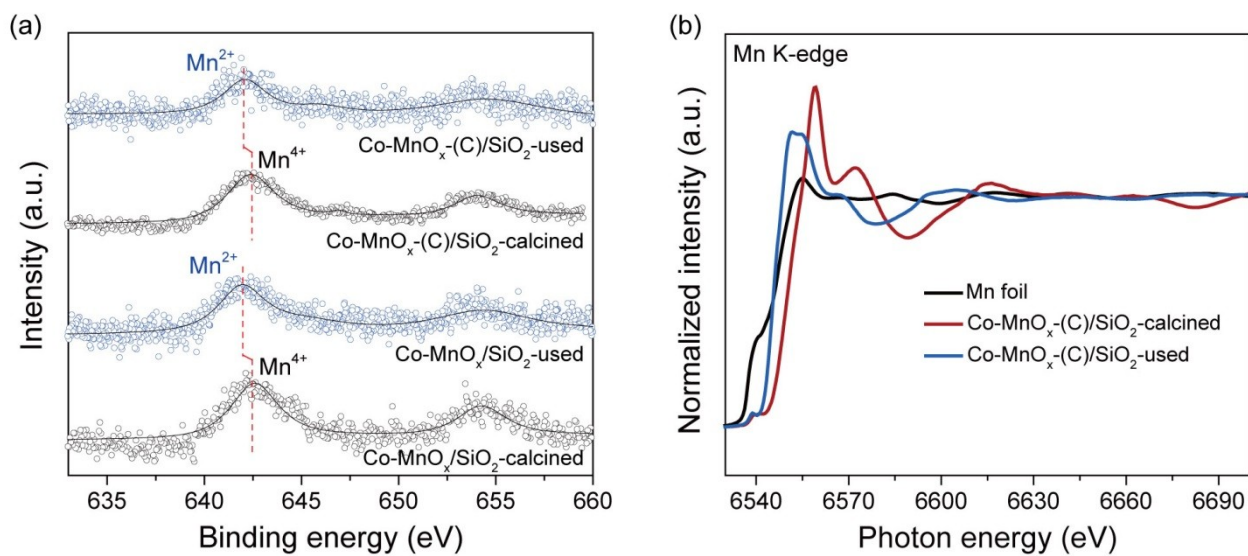
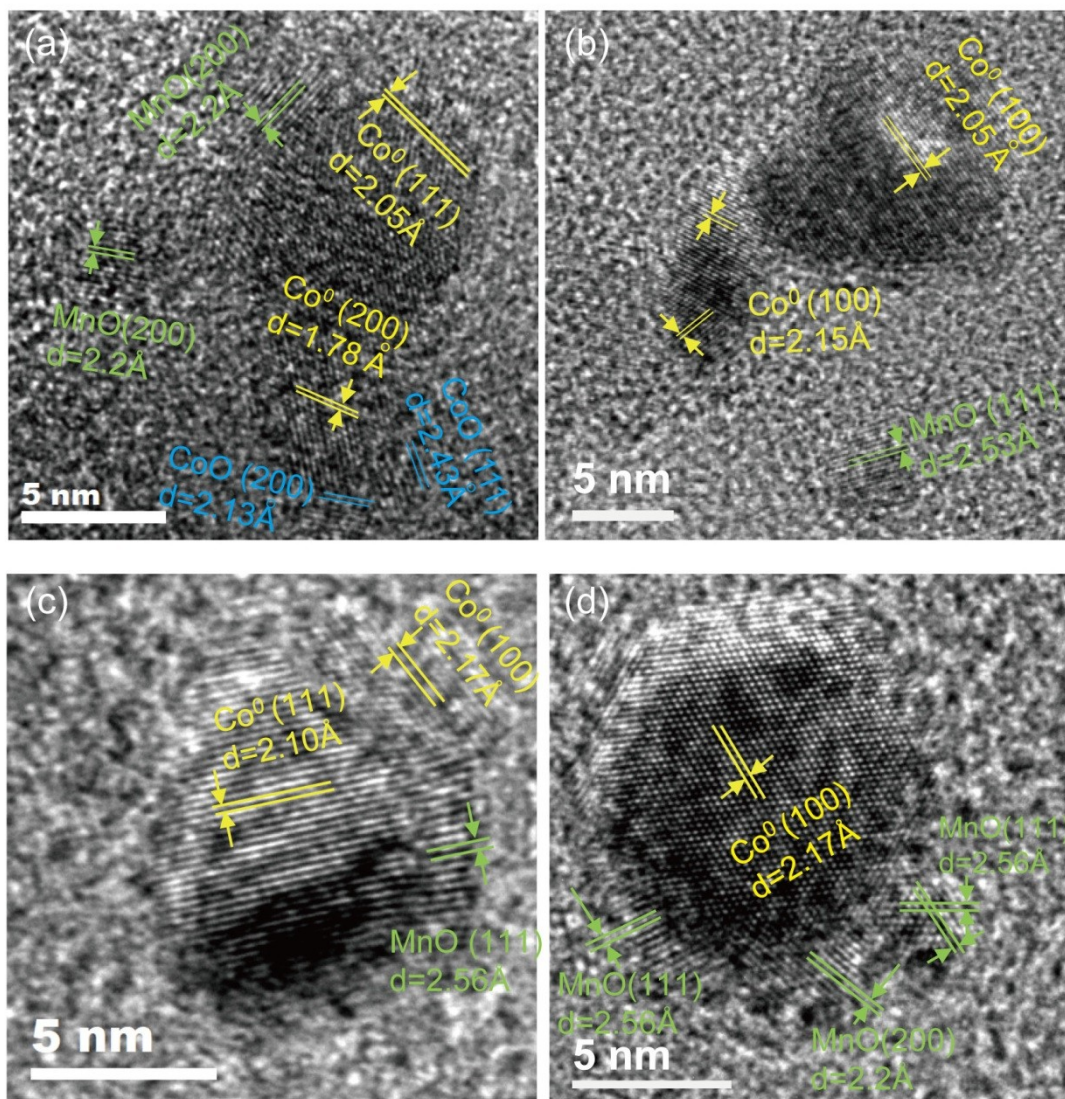


Fig. S11. (a) Mn 2p XPS spectra for the calcined and used catalysts. (b) Normalized Mn K-edge XANES spectra of the calcined and used Co-MnO_x-(C)/SiO₂ with the reference of Mn foil.



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280 **Fig. S12.** HR-STEM images of the used catalysts. **(a-b)** Co-MnO_x/SiO₂, **(c-d)** Co-MnO_x-(C)/SiO₂. The spatial segregation of
 281 MnO_x and Co NPs is clearly seen on the used Co-MnO_x/SiO₂. In contrast, MnO_x is found in a close spatial proximity with Co
 282 NPs on the used Co-MnO_x-(C)/SiO₂.

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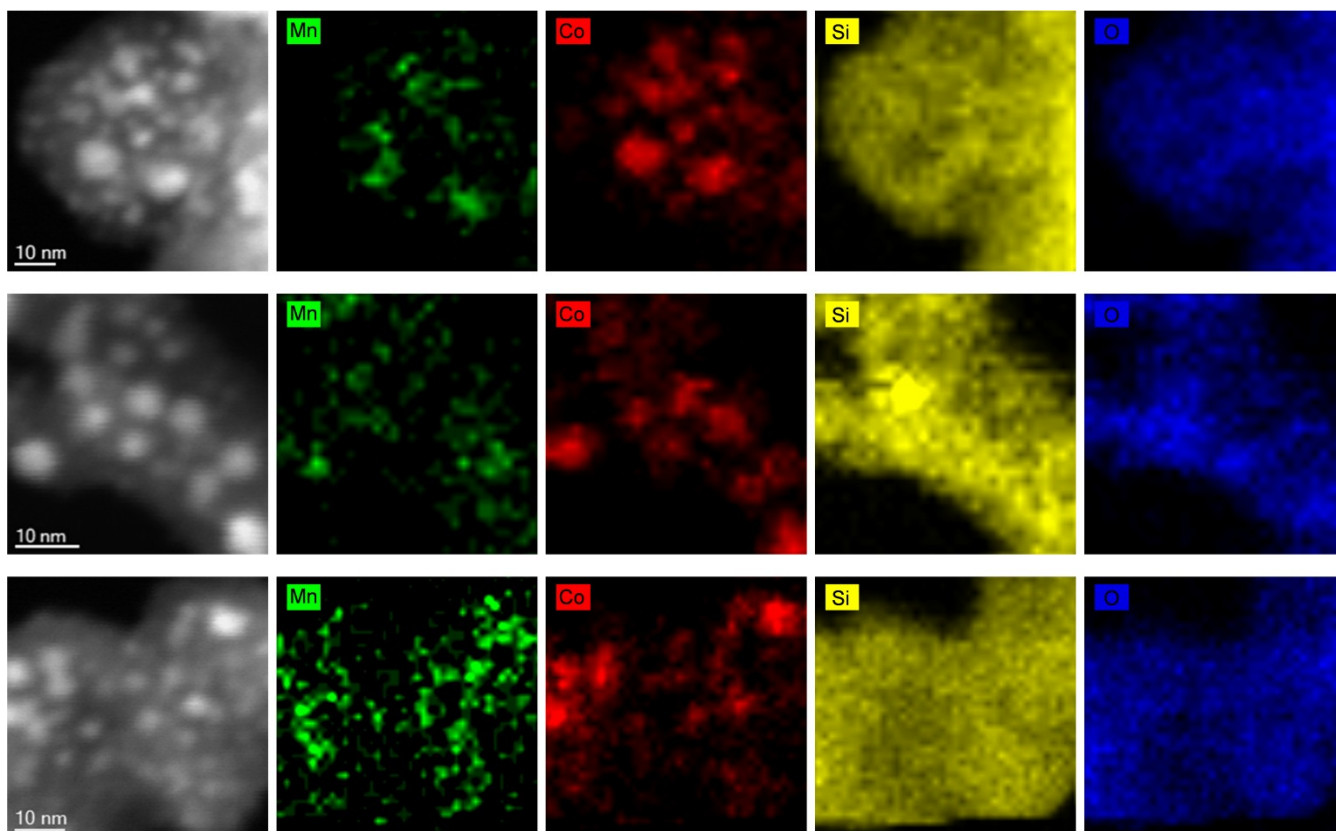
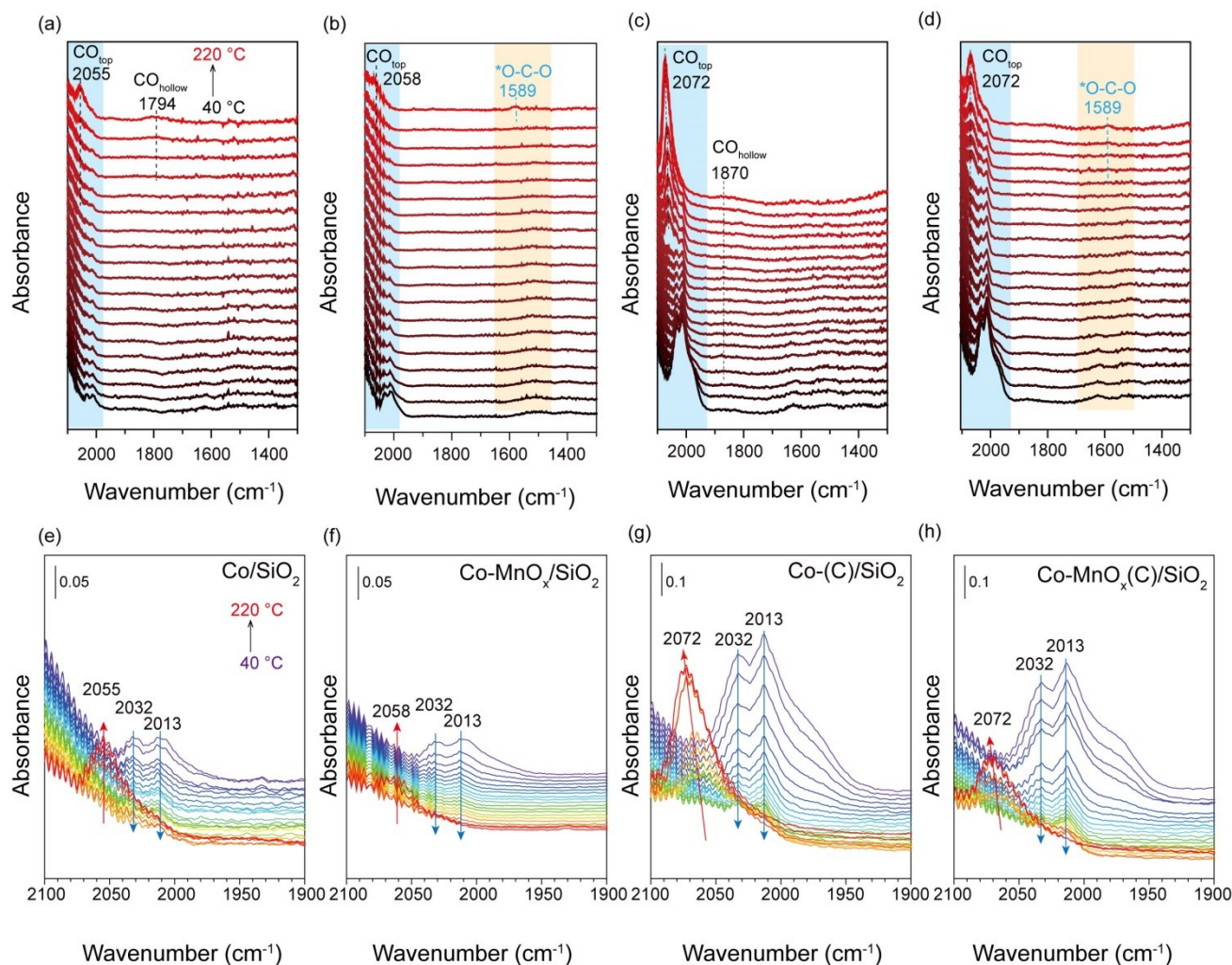


Fig. S13. High-magnification STEM-HAADF images on the random thin sample specimens of the used Co-MnO_x(C)/SiO₂ and the corresponding EDS elemental maps of Mn (green), Co (red), Si (yellow), and O (blue).



288

289 **Fig. S14.** *In situ* DRIFTS of CO as a function of temperature (40-220 °C) over the reduced catalysts: (a) Co/SiO₂, (b) Co-
 290 MnO_x/SiO₂, (c) Co-(C)/SiO₂ and (d) Co-MnO_x-(C)/SiO₂. Evolutions of CO_{top} with temperature in the wavenumber range of
 291 2100-1900 cm⁻¹: (e) Co/SiO₂, (f) Co-MnO_x/SiO₂, (g) Co-(C)/SiO₂ and (h) Co-MnO_x-(C)/SiO₂. The spectra were collected
 292 from 40 to 220 °C under the conditions of 20 mL/min flow of 30%CO/Ar at the atmospheric pressure.

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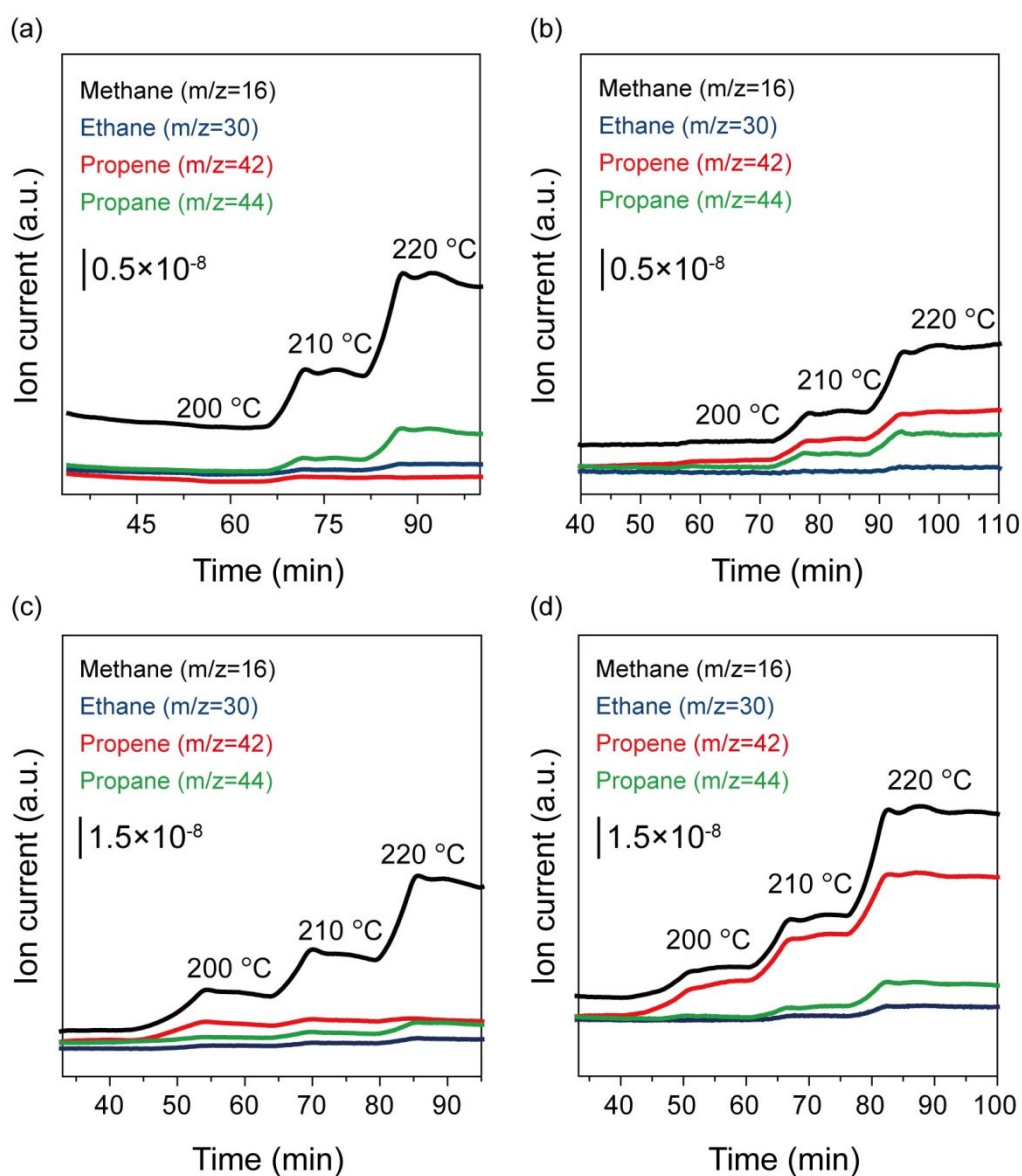
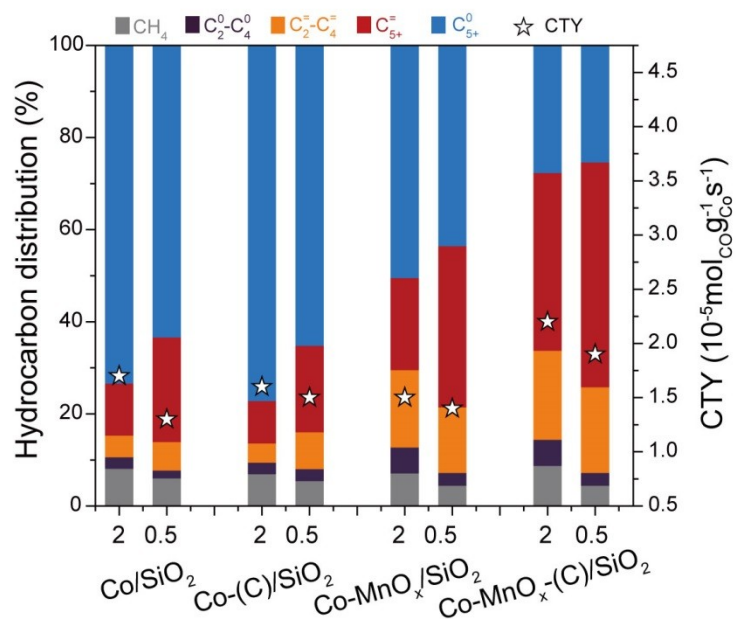


Fig. S15. The MS patterns of the gaseous hydrocarbons evolved during *in situ* DRIFTS of CO hydrogenation over (a) Co/SiO₂, (b) Co-MnO_x/SiO₂, (c) Co-(C)/SiO₂, and (d) Co-MnO_x-(C)/SiO₂.



298

299 **Fig. S16.** Catalytic performances of the catalysts at different H₂/CO ratio of feed gas. Reaction conditions: 0.50 g catalyst,
 300 1.0 MPa, 4.8 L·g_{cat}⁻¹·h⁻¹. In order to achieve the similar CTY for each catalyst, the reaction temperature is adjusted, e.g., 210
 301 °C for H₂/CO = 2.0 and 220 °C for H₂/CO = 0.5. The data was collected after more than 24 h on stream.

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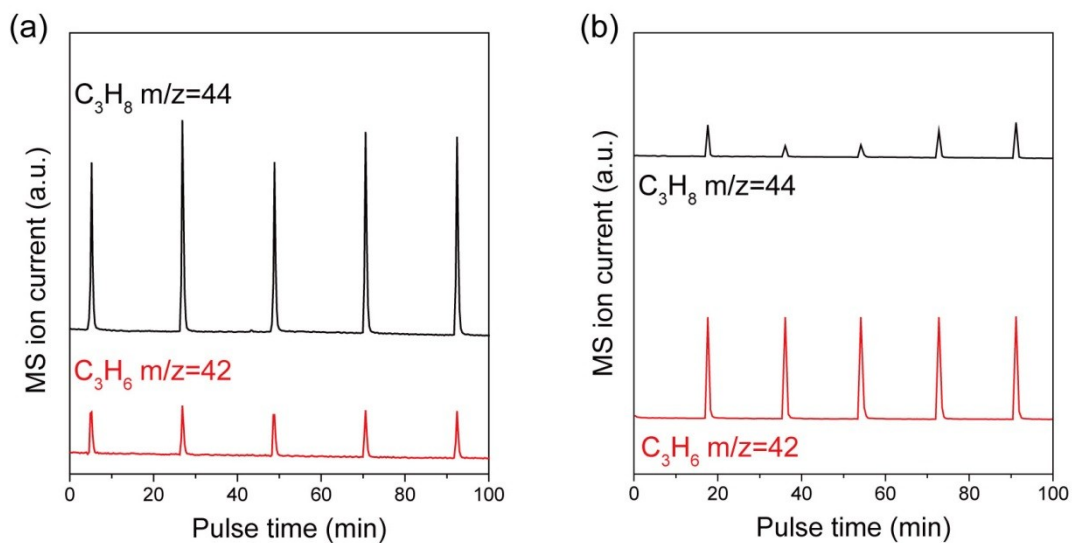


Fig. S17. Characterization of hydrogenation ability of (a) Co-(C)/SiO₂, (b) Co-MnO_x-(C)/SiO₂ on olefins by the experiment of pulsing propylene (20% C₃H₆ in Ar) into a flow of 10% H₂ in Ar. Before the pulse reaction, the catalysts (100 mg) were *in situ* reduced with 10% H₂ in Ar at 400 °C for 2 h, then cooled down to 220 °C. The C₃H₆ with m/z = 42 and C₃H₈ with m/z = 44 were detected by the mass spectrometry.

309 **3. Supplementary Tables**

310 **Table S1. Metal loadings and the calculated textural properties**

Catalyst	Co ^[a] (wt.%)	Mn ^[a] (wt.%)	Mn/Co molar ratio	Surface area ^[b] (m ² /g)	Pore volume ^[c] (cm ³ /g)	Pore size ^[d] (nm)
SiO ₂	n.a. ^[e]	n.a. ^[e]	n.a. ^[e]	223.4	1.11	19.9
Co/SiO ₂	13.0	n.a. ^[e]	n.a. ^[e]	180.5	0.80	17.7
Co-MnO _x /SiO ₂	12.9	2.5	0.21	211.8	1.06	19.7
Co-(C)/SiO ₂	13.0	n.a. ^[e]	n.a. ^[e]	209.5	0.95	18.3
Co-MnO _x -(C)/SiO ₂ -0.02	13.1	0.2	0.02	203.3	1.01	19.8
Co-MnO _x -(C)/SiO ₂ -0.05	13.0	0.5	0.04	212.8	0.94	17.7
Co-MnO _x -(C)/SiO ₂ -0.1	12.6	1.0	0.09	198.4	0.97	19.4
Co-MnO _x -(C)/SiO ₂ -0.2	12.8	2.9	0.24	201.4	0.89	17.7
Co-MnO _x -(C)/SiO ₂ -0.5	12.3	5.6	0.49	221.1	0.93	16.7

311 ^[a] Measured by ICP-OES.

312 ^[b] Calculated by the Brunauer-Emmett-Teller (BET) method based on the adsorption branch of the N₂ adsorption/desorption isotherms.

313 ^[c] The total pore volume determined by the Barrett-Joyner-Halenda (BJH) method at a relative pressure of 0.99.

314 ^[d] The average pore diameter evaluated by the BJH method.

315 ^[e] n.a.: not available.

317

Table S2. Catalytic performances over prepared catalysts

Catalyst	Catalytic activity		Product selectivity (%)					
	Conversion (%)	CTY $10^{-5} \text{ mol}_{\text{CO}} \cdot \text{g}_{\text{Co}}^{-1} \cdot \text{s}^{-1}$	CH ₄	CO ₂	C ₂ -C ₄	C ₅ +	C ₂₊ =	C ₅₊ =
Co/SiO ₂	28.7	2.1	9.3	1.0	6.0	83.7	8.6	5.1
Co-MnO _x /SiO ₂	35.6	2.6	7.2	0.9	11.8	80.1	16.7	10.2
Co-(C)/SiO ₂	32.2	2.4	8.5	1.2	6.9	83.4	10.5	8.0
Co-MnO _x -(C)/SiO ₂	76.8	5.6	7.0	1.8	12.1	79.1	21.1	14.3

318 Reaction conditions: 0.50 g catalyst, 220 °C, 1.0 MPa, GHSV = 2.4 L·g_{cat}⁻¹·h⁻¹ and H₂/CO = 2.0. All the reaction data is collected after
319 more than 24 h on stream.

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321

Table S3. Catalytic performances over prepared catalysts

Catalyst	Catalytic activity			Hydrocarbon selectivity (%)					$\alpha^{[b]}$
	Conversion (%)	CTY $10^{-5}\text{mol}_{\text{CO}}\cdot\text{g}_{\text{Co}}^{-1}\cdot\text{s}^{-1}$	TOF ^[a] (10^{-2}s^{-1})	CH ₄	C ₂ -C ₄	C ₅₊	C ₂₊ ⁼	C ₅₊ ⁼	
Co/SiO ₂	11.4	1.7	3.3	8.2	7.2	84.6	16.1	11.3	0.87
Co-MnO _x /SiO ₂	9.1	1.5	3.6	7.2	22.3	70.5	35.8	19.0	0.82
Co-(C)/SiO ₂	11.2	1.6	3.9	7.0	6.8	86.2	13.4	9.2	0.88
Co-MnO _x -(C)/SiO ₂	11.5	2.6	6.4	8.8	25.8	65.4	55.8	36.9	0.76

322 Reaction conditions: 0.50 g catalyst, 210 °C, 1.0 MPa, GHSV = 4.8~7.2 L·g_{cat}⁻¹·h⁻¹, H₂/CO = 2.0. All the reaction data is collected after
323 more than 24 h on stream. The CO₂ selectivity is ignored as it is less than 1% for each tested catalyst.

324 ^[a] The turnover frequency (TOF) was calculated based on the reaction results at an initial TOS of 1-3 h.

325 ^[b] The carbon-chain growth probability of α is calculated based on C₃-C₁₀ products.

326

Table S4. Comparison of catalytic performances with the typical Co-based FTS catalysts published in references for long-chain olefins

Catalyst	H ₂ / CO	Temp	Pressur	GHSV ^[a] (h ⁻¹)	CO	CTY ^[b] (g _{CO} /g _{Co} /h)	Selectivity (%)				STY	STY
		.	e		Conv		Selectivity (%)				(g/g _{Cat} /	(g/g _{Co} /
							CH	CO	C ₂₊ ⁼	C ₅₊ ⁼	h)	h)
		(°C)	(MPa)		(%)		4	2			C ₅₊ ⁼	C ₅₊ ⁼
1.5Mn10Co-SiO ₂ ^[2]	2.0	220	2.0	n.a.	10~15	1.6	11.0	<1	47.0	n.a.	n.a.	n.a.
CoMnAl@SiO ₂ ^[5]	0.5	260	1.0	4.0	12.5	n.a.	6.5	15.1	58.8	n.a.	n.a.	n.a.
2Mn-Co@C ^[6]	0.5	250	0.5	2.0	12.4	0.4	12.8	5.8	56.4	n.a.	n.a.	n.a.
2Mn-Co@C ^[6]	1.0	250	0.5	2.0	22.3	0.7	16.2	4.2	44.4	n.a.	n.a.	n.a.
15CoMn/SiO ₂ ^[7]	2.0	230	1.0	4.5	17.9	2.1	7.7	1.0	50.5	35.4	0.056	0.37
20Co5Mn/SiO ₂ ^[8]	2.0	230	1.0	4.5	25.7	2.3	7.1	0.5	44.6	35.0	0.080	0.40
Co@Co ₂ C ^[9]	2.0	220	3.0	2000 ^[c]	29.1	2.5	8.1	2.4	38.5	21.2	0.040	0.26
1.0PrCoRu/ Al ₂ O ₃ ^[10]	2.0	200	2.0	n.a.	20±3	1.0	8.4	0.9	31.7	19.9	0.015	0.10
Co-MnO _x -(C)/SiO ₂ (This work)	2.0	200	1.0	4.8	6.9	1.0	8.9	0.1	22.6	43.6	0.029	0.22
	2.0	210	1.0	4.8	18.6	2.7	8.5	0.5	50.1	32.5	0.057	0.45
	2.0	220	1.0	4.8	36.2	5.3	8.5	0.5	42.0	27.7	0.096	0.74
	2.0	230	1.0	4.8	47.1	6.9	9.0	1.0	31.8	22.7	0.103	0.79

^[a] GHSV, L·g_{cat}⁻¹·h⁻¹
^[b] CTY, 10⁻⁵ mol_{CO}·g_{Co}⁻¹·s⁻¹.
^[c] GHSV of 2000 h⁻¹ is used.
n.a.: not available.

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Table S5. Summary of the binding energies for different forms of Co from the deconvoluted XPS spectra

Catalyst	Calcined catalyst			Reduced catalyst ^[a]			Used catalyst ^[b]		
	Co 2p _{3/2} (eV)		Atomic ratios	Co 2p _{3/2} (eV)		Atomic ratios	Co 2p _{3/2} (eV)		Atomic ratios
	Co ³⁺	Co ²⁺	Co ²⁺ /Co ³⁺	Co ²⁺	Co ⁰	Co ⁰ /(Co ⁰ +Co ²⁺)	Co ²⁺	Co ⁰	Co ⁰ /(Co ⁰ +Co ²⁺)
Co/SiO ₂	779.1	780.8	1.8	780.8	778.4	0.36	780.8	778.3	0.52
Co-MnO _x /SiO ₂	779.0	780.6	2.4	781.4	778.8	0.35	781.4	778.3	0.77
Co-(C)/SiO ₂	779.2	780.9	7.4	782.2	778.2	0.04	781.1	778.3	0.25
Co-MnO _x -(C)/SiO ₂	779.4	781.0	7.9	782.6	778.4	0.03	781.5	778.6	0.51

335 ^[a] After reduction in 10%H₂ in Ar (10 mL/min) at 400 °C for 3 h.

336 ^[b] After reaction in syngas flow (32%CO/64%H₂/4%Ar, 10 mL/min) at 220 °C for 10 h.

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Table S6. Co K-edge EXAFS fitting results for calcined samples and the reference Co₃O₄

Sample	Scattering pair	CN	R(Å)	σ^2 (10 ⁻³ Å ²)	ΔE_0 (eV)	R-factor
Co ₃ O ₄	Co-O1	6	1.91±0.01	1.2±0.9	-0.97±1.4	0.0028
	Co-Co1	6	2.86±0.01	4.0±0.8	2.38±1.0	
	Co-Co2	6	3.35±0.02	4.3±1.3	2.38±1.0	
	Co-O1	4	1.82±0.04	4.0±0.8	-0.97±1.4	
	Co-Co1	12	3.35±0.03	4.0±0.8	2.38±1.0	
Co/SiO ₂	Co-O1	4.6±1.2	1.93±0.01	1.3±1.8	3.17±1.7	0.0012
	Co-Co1	5.5±1.5	2.86±0.01	3.4±1.5	3.17±1.7	
	Co-Co2	5.5±1.5	3.36±0.01	4.4±0.9	3.17±1.7	
	Co-O1	2.8±1.1	1.86±0.03	3.4±1.5	3.17±1.7	
	Co-Co1	2.8±1.1	3.29±0.05	3.4±1.5	3.17±1.7	
Co-MnO _x /SiO ₂	Co-O1	4.4±0.9	1.91±0.03	2.4±1.3	4.03±1.2	0.0009
	Co-Co1	5.6±1.1	2.87±0.01	4.0±1.3	4.03±1.2	
	Co-Co2	5.6±1.1	3.37±0.01	4.5±0.6	4.03±1.2	
	Co-O1	2.5±0.8	1.93±0.09	4.0±1.3	4.03±1.2	
	Co-Co1	2.5±0.8	3.33±0.05	4.0±1.3	4.03±1.2	
Co-(C)/SiO ₂	Co-O1	4.0±0.6	1.90±0.06	3.8±6.2	1.54±1.3	0.0006
	Co-Co1	3.2±0.8	2.84±0.01	2.5±1.4	1.54±1.3	
	Co-Co2	3.2±0.8	3.36±0.01	3.9±0.9	1.54±1.3	
	Co-O1	2.2±0.6	1.94±0.01	2.5±1.4	1.54±1.3	
	Co-Co1	2.2±0.6	3.26±0.12	2.5±1.4	1.54±1.3	
Co-MnO _x -(C)/SiO ₂	Co-O1	4.1±0.6	1.90±0.01	2.7±1.2	2.66±1.0	0.0004
	Co-Co1	4.4±0.8	2.86±0.01	4.4±1.1	2.66±1.0	
	Co-Co2	4.4±0.8	3.37±0.01	5.3±0.5	2.66±1.0	
	Co-O1	2.3±0.5	1.95±0.04	4.4±1.1	2.66±1.0	
	Co-Co1	2.3±0.5	3.31±0.03	4.4±1.1	2.66±1.0	

339 CN, coordination number; *R*, interatomic distance; σ , disorder parameter; ΔE_0 , energy shift. All the fitting analysis were performed in the *R*
340 space, $\Delta R = 1.0$ -3.4 and $\Delta K = 3.2$ -12.8. $S_0^2 = 0.59$ was obtained from fitting the Co₃O₄ standard sample.

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Table S7. The calculated reduction degree, H₂ uptake and particle size of different catalysts

Catalyst	Reduction degree ^[a] (%)	H ₂ -chemisorption			TEM	XRD	
		H ₂ uptake ^[b] (μmol·g _{cat} ⁻¹)	Dispersion ^[c] (%)	Particle size ^[d] (nm)	Particle size ^[e] (nm)	Co ₃ O ₄ size ^[f] (nm)	Co ⁰ size ^[g] (nm)
Co/SiO ₂	85.7	32.9	6.9	13.8	9.0	17.8	13.4
Co-MnO _x /SiO ₂	71.5	30.5	7.8	12.3	8.2	12.4	9.3
Co-(C)/SiO ₂	59.5	41.2	12.6	7.6	6.7	6.9	5.2
Co-MnO _x -(C)/SiO ₂	42.8	25.6	11.3	8.5	6.2	6.3	4.7

343 ^[a] Determined by the O₂ titration method.

344 ^[b] Calculated based on the H₂-chemisorption.

345 ^[c] Dispersion percentage calculated from $D = 1.179X/(W \cdot f)$, where X-H₂ uptake, f-reduction degree and W-the weight percentage of cobalt.

346 ^[d] Determined by H₂-chemisorption and calibrated by reduction degree.

347 ^[e] Determined from TEM images of calcined catalysts and calculated upon the measurement of random 300 particles.

348 ^[f] Calculated by Scherrer equation applied to the most intense diffraction ($2\theta = 36.8^\circ$) from XRD results of the calcined catalysts.

349 ^[g] Calculated according to $d(\text{Co}^0) = 0.75d(\text{Co}_3\text{O}_4)$.

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351 4. Supplementary References

- 352 [1] C. Márquez - Alvarez, N. Žilková, J. Pérez - Pariente, J. Čejka, Synthesis, Characterization and Catalytic
353 Applications of Organized Mesoporous Aluminas, *Catal. Rev.* 50 (2008) 222-286.
- 354 [2] D. Schröder, J. Thiessen, A. Jess, J. Scholz, Influence of the formation of short-chain olefins by manganese/cobalt-
355 catalyzed Fischer–Tropsch synthesis on the selectivity and effective reaction rate, *Catal. Sci. Technol.* 10 (2020) 475-
356 483.
- 357 [3] C. Erkey, J.B. Rodden, A. Akgerman, Diffusivities of synthesis gas and n-alkanes in Fischer-Tropsch wax, *Energ.*
358 *Fuel.* 4 (1990) 275-276.
- 359 [4] M. Zhang, M. Wang, B. Xu, D. Ma, How to Measure the Reaction Performance of Heterogeneous Catalytic
360 Reactions Reliably, *Joule*, 3 (2019) 2876-2883.
- 361 [5] T. Lin, P. Liu, K. Gong, Y. An, F. Yu, X. Wang, L. Zhong, Y. Sun, Designing silica-coated CoMn-based catalyst
362 for Fischer-Tropsch synthesis to olefins with low CO₂ emission, *Appl. Catal. B Environ.* 299 (2021) 120683.
- 363 [6] K. Gong, Y. Wei, Y. Dai, T. Lin, F. Yu, Y. An, X. Wang, F. Sun, Z. Jiang, L. Zhong, Carbon-encapsulated metallic
364 Co nanoparticles for Fischer-Tropsch to olefins with low CO₂ selectivity, *Appl. Catal. B Environ.* 316 (2022) 121700.
- 365 [7] J. Zheng, J. Cai, F. Jiang, Y. Xu, X. Liu, Investigation of the highly tunable selectivity to linear α -olefins in
366 Fischer–Tropsch synthesis over silica-supported Co and CoMn catalysts by carburization–reduction pretreatment,
367 *Catal. Sci. Technol.* 7 (2017) 4736-4755.
- 368 [8] Q. Lin, B. Liu, F. Jiang, X. Fang, Y. Xu, X. Liu, Assessing the formation of cobalt carbide and its catalytic
369 performance under realistic reaction conditions and tuning product selectivity in a cobalt-based FTS reaction, *Catal.*
370 *Sci. Technol.* 9 (2019) 3238-3258.
- 371 [9] Z. Zhao, W. Lu, R. Yang, H. Zhu, W. Dong, F. Sun, Z. Jiang, Y. Lyu, T. Liu, H. Du, Y. Ding, Insight into the
372 formation of Co@Co₂C catalysts for direct synthesis of higher alcohols and olefins from syngas, *ACS Catal.* 8 (2017)
373 228-241.
- 374 [10] K. Jeske, A.C. Kizilkaya, I. López-Luque, N. Pfänder, M. Bartsch, P. Concepción, G. Prieto, Design of cobalt
375 Fischer–Tropsch catalysts for the combined production of liquid fuels and olefin chemicals from hydrogen-rich
376 syngas, *ACS Catal.* 11 (2021) 4784-4798.